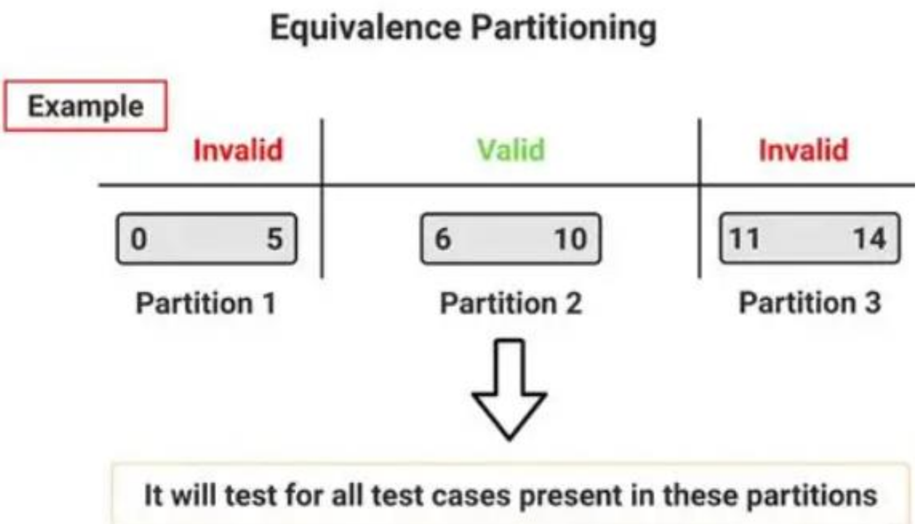
**Case Study 3:**

**Blackbox Test-Case Designed Techniques**

**1.Equivalence class partitioning (ECP)**

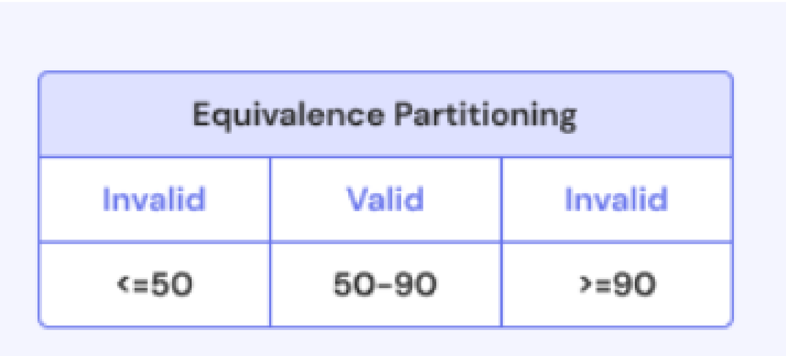
Design Equivalence Class Partitioning for an application number field that accepts 1-500 digits

**1. Range 0 to 14**



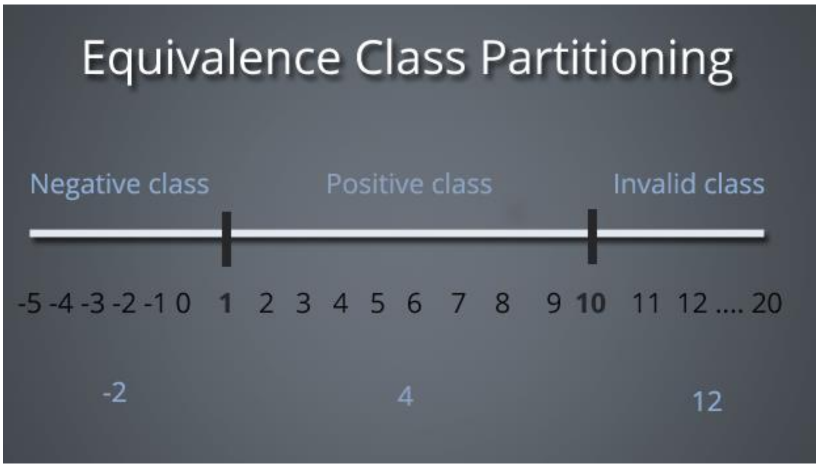

| S. No | Normal Test Data | Divide values into equivalence class | Test Data Using ECP |
|---|---|---|---|
| 1 | 3 | Invalid Class – 0–5 | 3 |
| 2 | 8 | Valid Class – 6–10 | 8 |
| 3 | 12 | Invalid Class – 11–14 | 12 |

**2.Range 50 to 90**



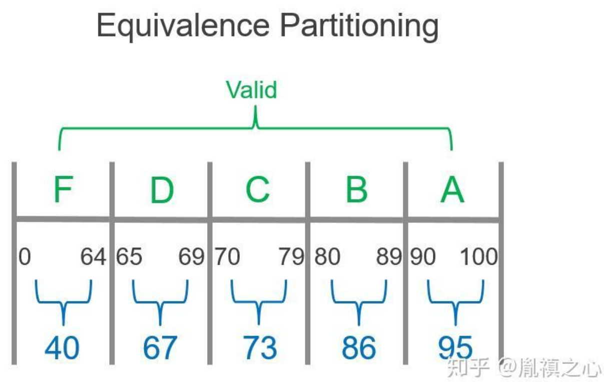

| S. No | Normal Test Data | Divide values into equivalence class | Test Data Using ECP |
|---|---|---|---|
| 1 | 40 | Invalid Class – <50 | 40 |
| 2 | 70 | Valid Class – 50–90 | 70 |
| 3 | 95 | Invalid Class – >90 | 95 |

**3.Positive number range**



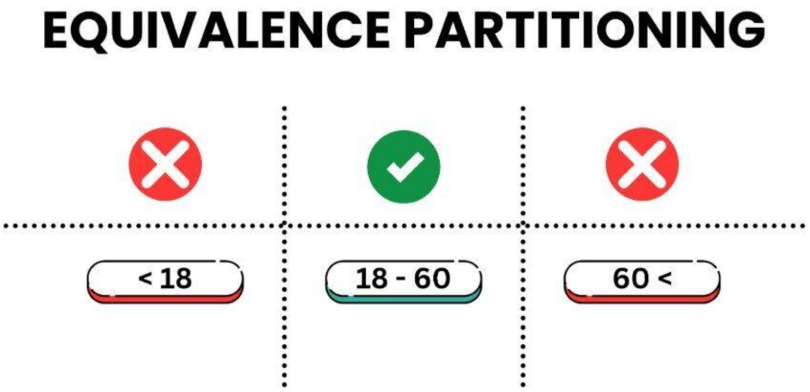

| S. No | Normal Test Data | Divide values into equivalence class | Test Data Using ECP |
|---|---|---|---|
| 1 | -2 | Invalid Class – Negative values (<1) | -2 |
| 2 | 4 | Valid Class – Positive values (1–9) | 4 |
| 3 | 12 | Invalid Class – Values ≥10 | 12 |

**4. Marks / Score Validation Using Equivalence Class Partitioning**




| S. No | Normal Test Data | Divide values into equivalence class | Test Data Using ECP |
|---|---|---|---|
| 1 | 40 | Valid Class – F Grade: 0–64 | 40 |
| 2 | 67 | Valid Class – D Grade: 65–69 | 67 |
| 3 | 73 | Valid Class – C Grade: 70–79 | 73 |
| 4 | 86 | Valid Class – B Grade: 80–89 | 86 |
| 5 | 95 | Valid Class – A Grade: 90–100 | 95 |

**5. Age Validation Using Equivalence Class Partitioning**




| S. No | Normal Test Data | Divide values into equivalence class | Test Data Using ECP |
|---|---|---|---|
| 1 | 15 | Invalid Class – Age <18 | 15 |
| 2 | 30 | Valid Class – Age 18–60 | 30 |
| 3 | 65 | Invalid Class – Age >60 | 65 |